

ActInf GuestStream #001.1 ~ Majid D. Beni "A critical analysis of Markovian monism"

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so uh hello i'm maggie oh we all all started all right so hello hi everyone welcome to the active inference lab and to the first active inference guest stream i'm here with our special guest majeed benny and we're just really honored and excited to be hearing your presentation we're going to be having a about 30 minute presentation on whatever you'd like to be speaking on and then i'm going to be collecting questions from the live chat and then relaying them to you and it was already just fun to talk to you right before we started so thanks again majeed and take it away go for it all right uh hello thank you very much uh yeah i'm very excited too and uh i thank you very much for giving me the chance to present a part of my work so uh i'm basically beginning to speak uh what i've done um a part of my work uh on a free energy principle so i will try to share my screen right now because i'm a great believer in extended mind and i barely could do anything without having my screen shared so you you mostly able to see my screen though yep it looks perfect thank you perfect thank you so uh uh i'm basically going to present uh a part of my work in a recent paper in centers critical analysis of mercury and monism and the structure of this talk is quite uh a straightforward so uh i'm not uh i briefly uh very briefly and with a very broad branch we'll speak about free energy principle and mercury and monism so i gloss over the technical stuff because for one thing i'm a philosopher and i'm not very well versed in these technical aspects and for everything i'm going to present the core of my work which is about scientific models uh so uh after that i speak briefly about scientific models and scientific modeling in philosophy of science and coming from the background of philosophy of science is what uh but it is the main part of my contribution to this literature linearity principle to look at it through the lens of philosophy of science and issue of scientific representation um then i'm going to um describe markovian models as some sort of scientific models and scrutinize their relationship with their target model in the real world and see whether any of these descriptions of the relationship between mercury and models and their target system could support smart coffee and monism and

briefly at the end i will also evaluate friston and colleagues attempted developing markovian moonism along the lines of structural realism so um so most probably you know more than a free energy principle and predictive processing uh the idea has been reintroduced to uh science but so the roots are this idea seems to be quite old but uh it it has been or introduced to resort to science by car freestyle and colleagues and it is all based on simple and subtle idea so uh generally it provides a framework that explains how a biological system exposed to random and unpredictable fluctuation in its external environment can restrict itself to occupying a limited number of a state and therefore therefore survive in some recognizable form so uh in order to uh be able to resist the uh dispersing uh force of the environment the organism have to in a nutshell have to keep the amount of its free energy low and uh predictive processing is some sort of theory of application indicating that when applied to the brain of some kind of system such as ourselves in all free energy minimizing free energy uh could be specified in terms of uh minimization of prediction error so there are these generative models in the brain and through a process of a top-down process the brain or the organism endeavor to uh apply these models to reality or at least to check them against real against the evidence that it receives from the external world um and then there is discrepancy it endeavors to uh uh somehow revise the models update its model or at least make changes to the environment so there is uh some sort of action by the covers mental action or over attacking but active influence is the uh other side of minimization of prediction error and uh princeton and colleagues mainly princeton um developed free energy principle into a fascinating theory of consciousness in terms of uh temporary richness or temporal depth of generative model so not only some organism can minimize their free energy and optimize the representation of the environment they could also somehow form some sort of a foresight or prediction about the possible outcomes of their action in the environment and it seems that this capacity is somehow tied up with conscious properties or consciousness and of course there is this idea of uh mark of blankets uh judah appeal introduced them i think in 1980s and there are some sort of bayesian networks uh which could help us to make a distinction between conditional independent account for the conditional independence between uh internal and external spaces so uh the literature itself uh free energy principle has been quite flourishing uh quite a number of scientists and philosophers uh have uh contributed to developing this literature and amongst the philosophers up to just to mention a few of them are great people who contributed such as uh jakub pauling andy clark sean gallagher and uh even people who are coming from a background of philosophy of science such as stefan hartman have contributed to this uh rich

literature uh but uh what what's interesting is that the free energy critical has not been developed into some sort of metaphysical test is about cognition or consciousness or brain so as far as i can see this uh paper which is the target paper of my paper incentives from cartesian duality to markovian modernism is the subtitle of prison and whis and hobson uh as far as i can see it is uh the main work the main paper uh that actually contributed to uh spelling out the metaphysical uh implications of the free energy principle this is the main paper as far as i can see that try to develop free energy principle into some sort of metaphysical uh testis of consciousness and i think that for this reason alone the paper deserves uh attention and uh i i'm a great admirer of the sport i usually do not engage with uh reviewing or criticizing single papers but this paper attracted my attention and interest because i think that uh this is so even before this uh people such as for for example fristen and thompson themselves try to develop a free energy principle into a theory of consciousness that provides a point to um cartesian dualism uh try to do so but as far as i can see here for the first time free energy principle has been developed into a very uh well-defined metaphysical thesis which is monism so the main point here is that i'm more sympathetic to the goal of this paper to the objective of this paper that at times the paper may indicate and what i'm going to say uh is mainly uh aim to invoke provoke the free energy principles to develop this markovian monism into a slightly a better uh account a well-supported account of consciousness metaphysical account of consciousness and i'm going to look at this issue through the lens of philosophy of science to see the relation between markovian blankets and other environments as some sort of relation between scientific models because markovian models are some sort of scientific models and i'm going uh argue for provide some argumentations for that uh to draw a connection between uh markovian models on the one hand and the target system on the other hand and so-called consciousness that appears through this relationship uh could be uh put into some sort of modernism which is a metaphysical test so i'm going to look at this issue science and unfold the implication as part of the implication for markovian monism so the test is itself markovian monism uh is quite a straightforward i have highlighted the part of it for you here so the thesis is is that fundamentally there is only one type of thing and only one type of irreversible property and markovian blankets form the calculus of belief which is not only a nice scientific mathematical model of consciousness but also provides a clear metaphysical statement and i can shed light on current philosophical debates so um the highlighted parts so it is this is the statement of markovian moonism and the red part indicates that oh how forcefully the theory have been uh spelled out in metaphysical terms so there is only one type of thing is a strong

metaphysical thesis and this is something that has been also stated in a Frieza and colleague's paper so and I here I highlighted another part of the same phrase which indicates that this statement of consciousness on the basis of Markovian models is not just a nice scientific or mathematical model of consciousness but also provides a metaphysical statement so the main question here is that how could we build Hong Kong that we could read of a metaphysical thesis on the basis of or scientific models of the phenomena scientific mathematical models of phenomena uh so I'm going to uh um ask this question as I say through the lens of philosophy of science to see what's the relation between I think that this basic question of philosophy of science what's the relation between scientific models and their target system they are significantly on the claim of Markovian monism which tries to develop some sort of a strong metaphysical thesis on the basis of this efficient modeling tool scientific modeling tool which are so this is an insight from a philosophy of science and the insight has been I think that originally it has its roots in works of uh Ronald Gearty and uh uh Paul Churchland for example uh and a number of and Bosnian foreign himself a number of important philosophers of science who have been developing this idea in a model-based approach to scientific tourism uh 1980s and it has been recently uh surfaced uh resurfaced in works of Michael Weisberg and uh Roman Frigg and uh God Frieza so and the general idea is that scientific practice is basically consists of model making making scientific models and the thing is that so if you have when engaging in scientific practice we just do not directly engage in in reading of the ideas from the feature of word uh reading of the theories the theories do not just latch on to the word directly there is some sort of representation and the representational relation between scientific theory scientific and the world is in indirect so scientific theories do not represent the features of the world directly the main thing here is that uh quite a few uh human factors interfere through the process of scientific representation so to name only a few for example scientist interests and explanatory uh goals and intentions uh play a role in how scientists are constructing their scientific models and how use them to represent the features of reality by scientific models and as uh well free for example among other people explains this you know process of indirect representation is quite sophisticated it includes several stages such as denotation or designation example exemplification kingdom and imputation so uh the main point here the take-home point here is that the relationship between scientific models on that one and our reality on the other hand or the target system on the other hand is quite uh sophisticated it is never straightforward and there are three ways of so and mark mark of blankets Markovian models are just efficient scientific tools in the

modelers of toolbox that allow the modeler to represent the features of the world in indirectly so i'm going to speak about three ways of uh understanding markovian models um i'm beginning so uh as the sketch that i showed earlier uh in the case i'm beginning with uh interpreting markovian models as physical or material models first and as fictional models and finally as mathematical models and this order has been chosen deliberately because i'm going from the at least likely reading of markovian models i'm going to are the most likely interpretation of markovian models so i'm convinced that markovian models could be understood this as some sort of mathematical models but i also consider other options because there are some sort of textual evidence to support them so for example we could it is indeed possible to understand more blankets or markovian models as some physical models material models for example christian says in uh uh 2019 paper that the marco blanket is not some aesthetic statistical device by which we could or we use it to observe or model the world it is the necessary attributes of the universe that can be carved into things so at least to a first approximation we try to read this phrase or the second phrase from the same paper it seems that uh for uh free energy principle theories for freedom for example a mark of blankets are actually features of the reality themselves they are physical things there are real real things out there and so uh i think that it is not uh only freezing is advocating this uh reading i think that also i i have seen similar phrases in for these and medsker's introduction to predictive processing and also in allen and freestone's papers since uh 2060 i think so uh this is there is some textual basis for understanding markovia but marco blankets as physical models but uh i think and i think that this indication this reference to mark of blankets as physical things has to be taken with a grain of salt i do not think that there are actually such things so if we for example open the brain or open an organic system i do not think that we can find anything resembling a markovian blanket there actually there is no blanket there there is no such physical thing but i can't understand if a terrorist uh could conceive of the target system as if it really include this uh representational device but the point is that even if we consider somehow considered that markov blankets markovian models are uh features of reality or physical things that exist in the world even if we can't say that it doesn't follow that reality itself shares the ontological and material features of markov blanket it doesn't follow that uh there is the same uh cells same ontological category that includes both markov blankets and their target system and there is an example that could demonstrate that so for example there is a such thing as a hydraulic pump model of economics so hydraulic pump is a physical object and for hydraulic model to work some properties of economic system mostly identified with

uh hitlering properties of the pond it could be for example uh exemplified through the process of imputation we could ascribe or attribute some of the properties of the literary poem to the physical object uh sorry to to ascribe some of the properties of israeli poem to the economical system that is supposed to be represented but it doesn't follow that the economical system and hydraulic pump which is the physical objects uh can be put under the same ontological category or material category so for example hydraulic foam which is a is made of for example steel or plastic or some some other material thing it doesn't follow that economy shares this material antique futures uh so uh i i simply uh put the uh materialist interpretation of mark mark of blankets aside and proceed to consider markov blankets as fictional models so and the main reason for this is that i assume that if we uh for example explore the actual brain of organism we couldn't find anything resembling a mark of blankets inside the brains so we could interpret them as some some fictional objects and uh so and the fictionalist and again there are textual support for this for example christian and colleagues say that if if we look into the behavior of creatures through the lenses of marconi and models it will look as if it is acting the creature is acting to minimize its particle or surf right zone so this as if phrase luke has a phrase uh i think that is very important it indicates that they do not really think that there is such thing as marco polanki's in the creature but uh it's look as if there is such thing um it has been this interpretation this fictionalist instrumentalist interpretation is gaining momentum as far as i can see so there are a number of other people such as maxwell ramstad and uh venus uh that are developing some sort of fictionalist and instrumentalist interpretation of free energy principle but the thing is that if you concept this fictionalist interpretation of marco blanket our free energy principle uh it follows that so if marco blankets are used for fictions we couldn't uh derive any sort of metaphysical conclusion from them so from a fiction so for example if you are studying uh greek mythology and assuming that greek mythology is some sort of fiction you cannot derive any conclusion about the nature of deity from what you read in a story from greek mythology there are no metaphysical conclusions to be derived from fictions and stories and this is precisely the point in invoking a fictionalized approach to scientific models so people are using fiction assuming that scientific models are fiction are useful instruments to just get rid of the metaphysical questions about the nature of fiction models or their relationship with reality so to make a longer story show this fictionalist interpretation doesn't lead to any sort of metaphysical conclusion about the nature of consciousness or the nature of mind or anything of that sort so this brings us to the last interpretation of markov blankets and it comes in terms of doesn't see how much time or everything i don't have much

time i try to wrap it up quickly so uh the the final is uh to interpret markov blankets in terms of mathematical uh and i think that there is good textual growth so uh it is to to at least to my mind to some extent from for someone who is coming from a background or philosophy of it seems rather obvious that marco collected are just mathematical modeling tools so they are mathematical tools in the context of bayesian networks to describe to represent their conditional uh independence between internal and external spaces uh and there is uh textual evidence for that again i have highlighted part of that which indicates that their enterprise aims to describe a construct from information theory and mathematical physics that helps them to define the properties of living system and their cognition and give them some sort of meaningful interpretation by invoking uh mathematical objects that could be used to consolidate this interpretation so uh and the use of so i i think that there is some sort of enlightenment inside uh in modern philosophy of uh in the wake of works of patrick soups and basman for us and in 1960s and 1970s uh so people's attention has been drawn the importance of mathematical models sec theory and model theory and that sort of stuff uh for modeling natural phenomena so when doing science scientists so the rise of modern in some sense the rise of modern science uh has been a result of invoking these sorts of mathematical approaches that allow us to represent and model natural phenomena of you know precision and vigor and rigor but uh the thing is that if we assume that markovian models are some sort of mathematical tools that can be used to represent the features of reality the thing is that we couldn't develop any sort of metaphysical conclusion about the nature of reality on the basis of application of these mathematical tools look if we assume that markov blankets are mathematical tools and if we want to identify their nature with the nature of reality the only conclusion that follows is that the reality is some sort of mathematical objects it's some sort of mathematical entity and interestingly enough as i will argue this is precisely the same sort of conclusion that uh that uh fristen and colleagues uh reached uh in the end of their paper so uh at some sense they are saying that they are in they intend to defend some sort of a reductive materialist or redacted physicalist but what happens is that they come to the conclusion that because these two ways of describing the nature of reality this division on the basis of markovian monuments to external and internal spaces because uh none of these interpretations known of these geometries that they develop none of this intrinsic and extrinsic perspective could be supported under the assumption that neither perspective is privileged one would have to conclude that reality is fundamentally ontologically neutral and they assume that by speaking of ontological neutral neutral nature of reality they are trying to build open uh the

mathematical nature of markovian blankets markovian to make concession on some sort of monism which identifies the nature of reality as neither physical nor mental uh but i would say that it would be quite a lot to read into the use of mathematical models so of course mathematical tools are neither in some sense are not neither a mental nor physical if you are not an idealist or realist especially if you are not a physicalist or idealistic mathematical entities but it doesn't follow that the target systems that could be used on the basis of using these mathematical tools itself the reality the target system that is going to be used by invoking this sort of mathematical tools itself is neither mental nor physical so this is the conclusion of the paper and for coming to that conclusion i have also discussed some sort of structural reality interpretation of the paper again it is about the relation between mathematical tools and the physical reality that they so uh again fristen and colleagues would be happy to uh make conversation on some sort of structural realities uh interpretation of their markovian modernism but in the context of the structural realism too a structural realist as some of the leading structural realists such as steven french and james lady man are quite explicit about this clear they do not make their ontological commitment on the basis of the mathematical framework that they use so they use mathematical framework they use modal theory they use set theory because they believe that this semantic framework modal theoretical approach or set theory capital to science uh virus it's uh structuralist loyalties on its sleeves but it doesn't mean that ontology itself could be read off from the features of this mathematical tools that they use so they are very clear that the job of the uh orthodox structurings are using mathematical tools such as set theory and modal theory and category theory but they are very clear that the job of these mathematical tools is mainly representational it is not constitutional it is not ontological on the other hand it is the physical reality that is supposed to be represented that has some sort of ontological constitutional rule and with regard to markovian monism if you want to derive any sort so the use of the mere application of doesn't lead to any sort of metaphysical conclusion because markovian models are themselves mainly representational tools they help us to represent features of the reality that is the target system the brain or or organism or the environment or whatever attitude it doesn't itself we could not build upon the features of this mathematical model itself to derive any source of ontological or metaphysical contribution of the features of the target system so uh thank you very much for listening and uh i i hope that what i said makes some sense at least to some of you but although i could not be completely sure so thank you very much awesome thank you for that fascinating presentation we already have a couple questions in the chat so if anyone else wants to ask questions we'll have a good

amount of time now just to talk a little bit and i'll just read um down the questions that were asked in the chat and a few things i wrote down so the first question in the chat was um as far as different styles of monism or different threads in the history and the philosophy of monism how does a dawkins based monism so thinking about like memes biological or non-biological mimetics that's kind of a dawkins monism how does that parallel or overlap with fristonian monism so in other words how does friston's monism or the markovian monism relate or overlap with other approaches to monist type theories uh well uh thank you very much uh for the question i'm so i'm not very knowledgeable about the different types of monies and so i knew about some quite classical theories of modernism i know a few things about for example russellian monism or uh the form of monism that has been developed by char sanders pairs aside from for example the classical forms that are developed by espinosa but so i'm not very well versed i'm not very knowledgeable about this kind of dark kinsey and momentism i barely know anything about memes and i'm not sure that we could identify men as real ontological entities but uh and i have already argued that i have some sort of uh misgivings about uh pristine and these uh and hobson's version of monism so uh i do not want to give a ready-made answer i would say that if the arkansas monism resembles a fristonian monism with regard to the fact that it involves memes which are uh as far as i can guess some sort of theoretical tools as features of reality uh it could be liable to the same kind of objections that i have uh experienced here against the fristonian kind of momentum thanks for the clear answer and i guess the follow-up on that is where do you situate this markovian in relationship to these classical kinds of monism that you just mentioned is that discussed in the paper or what how would you frame that so uh all right uh thank you very much again it is it is a fascinating question so before this i have developed some sort of a structural realist uh theories of uh consciousness too that happened basically based so my own approach to a free energy principle is associated with some sort of a structural realism too so again i myself have invoked some sort of mathematical structure embodied informational structures uh to describe the structure of the mind and consciousness and my basic uh argumentation there was that this structure sort of underpin the diversity between different icons of consciousness in terms of free energy principle uh gay org north hopes resting state activity theory and for example uh integrated information theory of consciousness so i myself have developed some sort of a structural realist but with regard to metaphysical aspects with regard to so i see monism specifically as some sort of metaphysical approach and also i have been very cautious with regard to stating a metaphysical stance so the version of a structural realism that i have

defended was some sort of epistemic structural realism it wasn't still realism but it was epistemic in the sense that i was ready to make commitment to the fact that to the extent that our scientific or scientific knowledge is informing us the future of reality is a structure but we couldn't make uh assertions about the nature of reality itself so whether it is a structure or not or whether it is uh the same stuff uh this the cell same stuff that is there or there are several kind of substances so uh i i profess agnosticism uh about the features of reality in itself so the version of a structural realism that i defended was episodic and i have no stance on the nature of reality in itself whether of monies or dualist types or any other kind of approach that's really interesting um seems pretty fair to be agnostic on the ultimate structure of reality and you had a quote on one of the slides that the structural realism is taking the pressure off of strong ontological commitments so how would you summarize this kind of best of both worlds structural realism approach are we just is it a buffet are we just taking what we like or what are we committing to with um epistemic structural realism very very very good so uh i think that it is partly an issue of motivations so uh when stating a structural realism for example people such as john laurel uh built open arguments from history of science to argue that so uh it takes the pressure of untold pressure in the sense that it doesn't make commitments to the intrinsic nature of entities it doesn't make commitment to objects it is committed mainly to the relations so uh and i think that uh i'm defending some sort of epistemic form of a structural realism because as i say it is completely agnostic about the features of reality itself so for example uh people such as john warrell are are defending structuralism on the basis of history of sciences with very clear examples for example from the history of optics to show that there is some sort of continuity between theories of optics although it seems that there are radical theoretical shifts in the history of science so some of the terms some of the theoretical entities that were supposed to be the subject of anthological commitments before are eliminated from or ontology today but it seems that there is some sort of continuity at the level of the structure of the theories at the level of form so they are not making come and in this sense they are indeed taking ontological pressure from uh their theory however there are other forms of for example anti-construction realism uh which itself comes in in various strides for example james layman is defending some sort of uh eliminativist uh approach the structural realism i think that for good reasons from the his from recent many particle physics uh but uh i'm slightly skeptical that so that sort of structuralism is taking any sort of in terms of ontological commitments i'm skeptical that it is less uh cumbersome that the orthodox forms of uh scientific realism i'm sure that there is a perfect match between elite eliminating structural realism and what we know

about what modern physics many particle physics is representing about the features of the world but with regard to metaphysics for example antics structuralism claims that all that there is is a structure which is a strong metaphysical claim so it eliminates entities objects from the ontology which is quiet ambitious and a strong ontological commitment so and the thing is that i do not think that the version of structural realism that freestone and colleagues have defended is epistemic because they are defending some sort of metaphysical thesis here monism is some sort of metaphysical thesis so although structural realism was supposed to be ontologically modest in the sense that at least for example world's version doesn't make commitment to the nature of the world itself the metaphysical versions of a structural realism such as james lady are including are harboring strong metaphysical tests about the features of reality that they are claiming to be completely structural and i think that in this sense markovian monism that is defended by frieza and others is not metaphysically modest at all and i think that this is one reason so i didn't and thank you very much for bringing this point to my attention i didn't expel this point clearly in the paper because i didn't explain many times to discuss the structuralism this is specific paper but i think that not as i look at it from uh here i can see that it could be the main motive behind my mind that encouraged me to be a bit critical about this form of uh monism that is uh presented in this paper wonderful paper by fristen and colleagues cool cool response here's another question from the chat kind of keeping with this theme of physics and realism and scientific communities that are evolving their models you mentioned optics so let's think about particle physics cambridge breaths wrote can the author describe in his personal belief the main difference between a mathematical markov blanketness of an electron when it's behaving as a particle versus when it's behaving as a wave so how do we think about quantum mechanics and quantum phenomena when we're talking about these different realism ideas well to be quite honest uh i have no idea do you think that it is mainly because of my ignorance of uh modern physics so i have some general conception of what happens at the level of many particle physics but how to state that in terms of marco and i can imagine how could we state that in terms of markovian blankets but i cannot see how the distinction between a particle being in a vagus state or being an entity being in a very state or participant uh could be related to this issue of markovian models i assume that markovian so one thing as i know about markovian models is that they are so powerful they are such powerful modeling tools that they could be used almost to model everything that there is to be modeled uh but well and i uh suspect and it is only some sort of uh wild guess that we could model the behavior of an entity either in in its their state or as far as we try to provide some

sort of probabilistic representation of the behavior of the entity to model the conditional independence between different states of identity we could involve markovian blankets marco blankets to do so but i think that at the same time i miss the main part of the uh the core of the question i think um just maybe guessing from the context the electron is something that we might say we might want to say it ontologically exists or some might think it ontologically exists yet it can display this sort of bimodal behavior when it can depending on the double slit and the context of the observer it can be acting yeah great so all right so how does a markovian blanket does that so uh uh as i say this this kind of approach to the behavior of entities in many particle physics was what motivated people such as steven french and james ladyman to develop forms of structural realism so it is not only an issue of an electron the thing is that when we look at this uh the behavior of entities at the level of at the fundamental level if i don't want to use a fundamental at the level of many particle physics they should do all behavior there is some sort of package that includes individual objects and it is the interpretation for example of the electron that allows us to identify that as some sort of individual objects and there is a package that doesn't allow us to model it as an individual object so there are both so we have to conceive of that as both individual individual objects and non-individual objects and this seems to wreak havoc with your ontology of modern physics and structural realism provides an answer to that saying that all right let's consider the common reality let's consider the common structures that underpin both formulation and the behavior of electrons for example are stated in several sort of statistics that are developed from maxwell's war for example both einstein statistics and other sort of statistics so i assume that we could also invoke some sort of a mark of blankets to behavior to to model the behavior of this kind of entities at the level of subparticle physics but uh does the question mean to indicate that there would be some sort of ontological gain to do so uh the thing is that even in philosophy of physics the the discussion of a structural realism whether we could invoke some sort of representational mathematical tools or we have to just rely on the geometrical or physical statement of the situation so um i think that these are sophisticated issues in the uh philosophy of physics and i'm hardly an expert in that field uh i'm not very clear but uh obvious advantage with a state of regimentation of a state of subparticles in terms of markov blankets would do to the extent that metaphysically called stencils or attributes are a christian or attitude cool so here are a few um general questions what does it mean to be ontologically neutral so that was brought up and i was just wondering what what is the alternative or how could it be charged or not neutral what is that there are several ways of reminding

ontologically neutral so for example uh a way that i prefer is to as i say process ontological agnosticism to not make any commitments about the nature of reality itself but i do not think that it is this sort of ontological neutrality that is at issue in freestone and colleagues paper i think that they want to be ontologically neutral in the sense that so this is this traditional uh division in the philosophy of mind between physicalism between dualism for example physicalism people who think that the nature of mind is physical and uh mind or at least mental processes are different from physical properties so i think that markovian monism wants to be it doesn't want to make commitment to either mentalism or physicalism it doesn't want to identify uh either in physical terms or mental terms and in this sense it strives to remind ontologically neutral which i think it is quite cool so as i say i'm a great admirer of this project because i think that it is the first significant attempt that tries to develop free energy principle into a remarkable metaphysical testis and i think that if free energy principle theories could really show that the theory could support or could be construed along the lines of neutral harmonism to show that it really supports a stance that doesn't commit us either to conceiving of the nature of mind as a mental state or as a mentalist or mental properties or as physical it will be a real achievement i think uh but at the same time i think that just depending on the mathematical uh framework of markovian models doesn't provide sufficient purchase for doing that it doesn't provide a sufficient basis for boosting a well-supported form of monism very cool so it sounds like there's the eternal debate of whether mind leads to matter or matter leads to mind is there one type of thing two kinds of things and it would be awesome for free energy principle to step into that gap but what we've seen so far with the of the mathematical construct of the markov blanket does not adequately resolve this metaphysical question so what kind of analyses or research or measurement or calculation i mean i'm a scientist so i'm thinking about measurement and experiments but as a philosopher what would um reduce our uncertainty about where the fep fits into all of this so uh far far it be from a philosopher to try to prescribe or endorse anything the scientists the thing is that i feel that so so a cool thing about cognitive sciences neurosciences is that i think that uh philosophy car ideas could really be useful to scientists in this field but uh the nature of this collaboration the nature of what happens here is so sophisticated that one can hardly so so i'm told that prediction is pretty hard especially about the future i i couldn't have any idea or any prescription what can we do to bring to to make this approach more fruitful the only thing that we could i think we could do is precisely the thing that we are doing just trying our best and uh collaborating so and i think that this is why this active inference lab is a cool idea because it allows people from different

background come and share their ideas and we will see whether anything uh significant will emerge from all of that or not but i think that this is the only thing that we could do oh it has been the only thing that we have been doing for quite a while and paid well those are cool very it's almost like there's the mind matter dualism and then there's the science and the humanities or science and philosophy and so is one going to envelop the other is there really only one and the other one is kind of a sub field and instead of playing into that dichotomized debate which is polarized it's almost like we're taking an ontologically neutral stance here and highlighting the importance of coming together across different contexts and having that conversation so my question on that would be let's just say that there was somebody who were curious and wanted to jump in to learn more what kind of research or skills or mentorship might help this person on their journey whether they're you know from a philosophical side or from a scientific side what do you think would be some key tips for somebody who's just starting on their fep journey uh well um the best thing that i've heard about the relationship between these different disciplines comes from a stephen french who is a structural realist and also a wise philosopher of science saying that metal physics is best when informed with science and science is best then uh i don't want to say that informed by metaphysics but that is taking some metaphysical insight into consideration so but on the other hand i think that the nature of uh scientific and academic work these days is so so in other words the scientific activity is so compartmentalized these days people are doing their different works in different disciplines in their own labs and uh the nature of the problem is so hard that they have to spend all of their time on specific problems so uh i can hardly also provide any advice of what kind of mentorship suits anyone who wants to work hard up do some excellent job in frener's principle what i can say that what's very interesting about free energy principle is that it seems to provide at least some uh purchase for for a discourse some space for a discourse between scientists and philosophers so it seems that some of us some of the uh good scientists such as for example maxwell ramstad or uh alex constant or fristen himself who are scientists are are completely interested in philosophical discussions of what's happening here and it seems that what's even more wonderful it seems that uh philosophical discussions are affecting the nature of their work in some sense i think that it is something that you can hardly observe for example uh age wr theories of physics these days uh i don't know whether the great physicists are studying philosophy for their amusement or not but i doubt that they are in this sort of constructive uh relationship with uh uh i think that this kind of and the nature of free energy principle itself is quite unificatory brings different disciplines because uh ideas from different

disciplines together so i have no clear idea about the or any kind of schedule or program but i can say that people who are interested in free energy principle and studying that have already chanced to get a wider scope with wide intellectual scope with insights from a number of different disciplines biology computational neuroscience uh cognitive psychology philosophy of mind uh philosophy of biology and that sort of thing so it seems to come with the discipline as it is i [Music] and i liked how you grounded the paper and the presentation in the history and the philosophy of science because it is a community that enacts the conversation around the ideas and then when there is not a healthy dialogue it's kind of like the philosophers are uh sweeping up after the scientists or vice versa or they're just sort of spectators to each other but then we're almost moving towards a a fruitful bi-directional conversation that actually as you're pointing out with the specific people who are taking this into their research approach it actually is something where the science and the philosophy are co-influencing each other so that's very interesting and special it's almost like it's such a big scientific idea and so challenging that we know that we need multiple expertises we know that we're gonna need to be working on it together for a long time then that helps set the stage for something that might surprise people because it's not just coming from their camp so we have history and philosophy science and then i guess another thread is math mathematics markov blankets as you pointed out came from pearl in the 1988 work and has been a lot subsequently where does set or category theory come into this uh well well uh very good question i think that in some sense it depends on your view of your your perspective on mathematics if you are set or taking uh foundations in mathematics if you are a mathematician who uh prays to the god of set theory every night in his loneliness you naturally think that every kind of mathematical theory is coming is based on sectarian security has a general foundation or there are other people who are ready to give this rule to the category theory but uh for me because my uh engagement with these ideas come from a background of structural realism the main thing is that uh and uh it has been going to the wonderful works of uh soups uh patrick soups and van frozen philosophers of science came to so so rational reconstruction have been a program from the very beginning of the history of philosophy of science so philosophers and again it is why free energy principle is so interesting for me because because of its great unifi unificatory scope and there have been philosophers of science and there are still philosophers of science who think that the main job of philosophy of science is to form some ideas about unification of sciences and in order to bring about this unification uh philosophers thought of using logical formal frameworks for state regimenting different theories from different disciplines so for

example in physics people are using a special form of mathematics in biology when they are using mathematics at all they may use a statistical uh theories or different forms of mathematics and they came to this idea if that if we use some form of logic some form of for example set theory model theory we could have a cool mathematical framework that allows us to regiment different theories from different disciplines and clarify their relationship together and have a realistic clear neatly stated view of what's happening in the world assuming that science provides a reliable picture of the world if we could use model theory or set theory and the thing is that uh and i stated this view in a book that has been published by spring you're according to a structural i begin to have doubts about the usefulness or at least the realistic impact of using set theory and model theory to represent the features of the world so i become a skeptical in the sense that i thought that this uh set theory and model theory uh frameworks are too abstract and even if we could regiment the structure of sciences into these frameworks we still couldn't account for scientific representation because these are the set theory and category theory do not represent the word these are scientists who are representing the word using mathematical models so i try to depart from the usual ways of scientific representation on the basis of sectarian model theory and invoke other sorts of mathematical frameworks that could uh provide a better match a more congenial tool for accounting for scientific representation it was very beginning to uh try to introduce some sort of information theory framework and not only information to article anymore but embodied information sort of framework meaning precisely the same sort of framework that fristen and colleagues are using in approximate uh bayesianism to regiment the structure of scientific theories i came to the idea that it allows us to have a more realistic account if we be able to regiment the structure of theories into this approximate bayesian framework for example instead of say theory and modal theory we could provide a better understanding of the relationship between scientific theories which are actually embedded in the brain of scientists or at least in the scientific community and their target system in the world because uh uh their set theory and model theory are too abstract to take care of the job uh this kind of approximate bayesianism is something that at least according to our best scientific theories is by default embodied in the brain of organism and in this sense we could uh have scientific representation super window the relationship between the organism and its environment but it is another story it is not directly related to this markovianism thing that i developed here well it is it's super interesting because this embodied and enacted relational approach we can't leave the scientist out of the picture so it's almost like we we remove too much when we abstract that relationship away

and then it almost hints towards that electron question from earlier which is well if your ontology or your model is just an electron not even the context you're going to be lost because you won't understand why certain things are happening to your system of interest and so how do we pull back rigorously and include enough context setting so that there is an ontology or a local logic that's making sense so it's just cool like how the historical thread the philosophical thread it's not um spice or salt on the research like just adding it up it's actually something that is um intertwined and and sets the total scope of the scientific journey so it's like very cool to hear about that so if there's any other questions in live chat we'll ask but yeah only with regard to a quick follow-up about the last point that you mentioned so part of that might be the contemporary philosophy of science at least the dominant approach which is structural realism is that precisely as you mentioned it leaves those human beings out of the picture when it comes to scientific representation so uh there were a bunch of brilliant philosophers of science such as ronald geary and for example nancy narcissian who is fortunately still active who try to provide some sort of cognitive model of scientific practice itself of scientific activity but for some reasons i i'm not sure about the reason it has been pushed aside uh and it is not the dominant view of philosophy of science again so part of my activity in the philosophy of science was to bring this sort of cognitive approach back to the foreground and i think that i was fortunate that i bumped up to this free energy principle and predictive processing because in order to account for cognitive models of science so we uh sometimes we do not just want to sit in your armchair and do philosophy of science but want to support our philosophy of science by some actual scientific theories and in order to develop these cognitive models of science cognitive approach to philosophy of science it seems that free energy principle and predictive coding are fascinating tools because they allow us to provide a very good well supported picture of the relationship between scientists who are basically organism and the community of scientists uh which could be uh specified in terms i don't know free energy principle common theories of communication under the relationship between community and scientists on the one hand and their environment the world on the other and i think that this picture is much more realistic than the ordinary but the orthodox picture of scientific that is fashionable yep cool it's like a yes and with action and inference so philosophy isn't just the armchair what does it look like to be embodied and inactive and then perhaps it looks like what is known by the career name of science but we know that learning and research happens inside of academia outside of academia so it is a more realistic picture and here's a related question from the chat stephen asks this ability to dissolve the internalist and externalist

duelist position is a powerful part of markovian monism thus giving an open dynamical system uh uh the adaptive open boundaries any thoughts on this yeah i completely agree in the beginning i told that i'm a great admirer of markovian monism i think that it is a bigger step forward and this is precisely because of this reason because it goes beyond this kind of orthodox dualism between intel so here it is described as some sort of dualism between internal spaces and external spaces and um frison and colleagues put a lot of force and energy into developing this kind of informational geometries that again i think is a wonderful uh formal framework to model this idea so this is precisely uh in my opinion for what it is this is precisely the valuable part of markovian modernism that it tries to use this kind of scientific that goes beyond this dualism between internal spaces and external spaces to develop some source of uh uh some sort of hormones and metaphysical tests so i completely agree that this is a brilliant idea so the question is what source of philosophy called argumentation or scientific argumentation could be offered to support this picture so i think that's as it is the status quo of this theory is not strong enough to carry the metaphysical weight the question is and as i said earlier the paper uh aims to come with the hope to provoke the free energy principle theories to develop this idea to develop markovian modernism so what sort of philosophical or scientific argumentation could be offered to uh support this insight that has been mentioned by stephen and lies at the core of markovian mode this is the main question awesome so just in the last um few minutes here this paper came out you know one week ago you probably wrote it several months before but what are your next research questions or what are some exciting ongoing threads like where do you go let's say someone read this paper where are you taking this frontier of thought uh all right again uh oh cool idea so uh so as we have already gathered uh my mind is very disorderly and it is because i'm cursing several lines of research together so uh part of my research right now is to develop this kind of model-based understanding of markovian monism to defense of modularity tests it's not precisely the source of fedorian modularity but at least the source of modularity that are defended by for example the alcohol and i think that the model-based approach provide at least some some grounds for defending this source of modularity this is on the other hand i'm trying so uh the other thing is that i feel that uh there is ample room for good metaphysical work on this free energy principle and free energy principle theory of consciousness so uh as you may know there are already so markovian monism for example offer some sort of metaphysical realist reading of uh theory of consciousness but there are also illusionists constitutes illusionist interpretations a theory of consciousness under free and i think that for some reasons none of this provides a quiet good

metaphysical interpretation theory of consciousness under free energy principle so i'm developing another theory i'm trying to develop a theory of consciousness under free energy principle that could be construed along the line of uh strong intentionalism and it is associated with uh works of historical works of uh friends barantano and tinkering more recently so uh it is another line of research that i have in mind to associate this theory of consciousness with some sort of intentionalism uh and then again uh i'm engaged with uh as as the paper may indicate i'm looking for further groans for for the reasons to develop or at least support this markovian monism myself so as i say this paper is just the first the negative part of some sort of argument that has to be produced by others or myself so uh i also i'm trying to because i because i can't see the appeal of this markovian monism so i try to find reasonable groans for supporting this theory i'm not sure where does it go but yeah it is an awesome approach a lot of times a critique it's almost like consequentialism someone says i don't like your outcome so i'm going to look for all the details or say that it wasn't really supported and you're almost like saying actually i do see that goal or i see the benefits here and we just need to navigate in a slightly new way but we're going to be learning and connecting these ideas along the way because it isn't mapped how we're going to get to markovian monism and maybe that's not where we're going but we're working towards brilliant i get excited and caught you because it is related to another part of my research which tries to relate free energy principle to group group knowledge and uh to account for for example collective intentionality on the basis of free energy principles so uh freestone and fears have produced a very fascinating theories of communication and social relation under and i think that i think that this theory could be applied to develop the part of the epistemological social epistemology and i think that there would be wonderful wonderful reasons so uh it is natural for me to see this kind of scientific back and forth between different uh intellectuals as some sort of uh collaboration that consequentially consequentialist as you phrase that masterfully that hopefully leads to the growth of knowledge in general yes the individual in the armchair it lends ones toward it lends ones almost to neuro reductionism and individualism but when the person is walking around and at a conference and they're they're sharing their experience and they're learning and being surprised by by friends and colleagues then there really is something that's more like distributed cognition so i'll really look forward to seeing that um majin thanks so much for this appearance this was a epic and special first guest stream for our lab and you're always welcome just let us know if you have any other topics that you want to come on a stream with you or colleagues or students or anything we're always there to host uh and hold the space for

this kind of work so again thanks a lot for the research and i'm sure everyone really appreciated hearing it thank you very much for having me and also cool work to to do this uh active inference lab so uh good luck with that ah and very well done and thank you very much for having me great we'll be in touch so see you later